SQL

```
mysql> desc customers;
+-------+--------------+------+-----+---------+-------+
| Field | Type         | Null | Key | Default | Extra |
+-------+--------------+------+-----+---------+-------+
| email | varchar(255) | NO   | PRI | NULL    |       |
| fname | varchar(100) | YES  |     | NULL    |       |
| lname | varchar(100) | YES  |     | NULL    |       |
+-------+--------------+------+-----+---------+-------+
```

SQL is based on table and column names

SQL is not case-sensitive

`select * from customers`

`select * from customers where fname = 'Raj'`

`select email, fname from customers`

JP-QL

```java
@Entity
@Table(name="customers")
public class Customer {
    @Id
    private String email;

    @Column(name="FNAME", length = 100)
    private String firstName;

    @Column(name="LNAME", length = 100)
    private String lastName;
}
```

JP-QL is based on class and variable names

JP-QL is case-sensitive wrt to class and variable

`from Customer`

`from Customer where firstName = 'Raj'`

`select email, firstName from Customer`

Mapping associations:

1) One to many
2) Many to one
3) Many to many
4) One to one

```
mysql> select * from customers;
+-------------------+--------+-------+
| email             | fname  | lname |
+-------------------+--------+-------+
| geetha@adobe.com  | Geetha | Mohan |
| rita@adobe.com    | Rita   | Jones |
| roger@adobe.com   | Roger  | Smith |
+-------------------+--------+-------+
```

1

*

orders

Oid	Order_date	Customer_fk	amount
343	22-JUL-2025 4:50:11	rita@adobe.com ✓	453535
344	22-JUL-2025 6:10:10	roger@adobe.com	64234
345	23-JUL-2025 10:30	rita@adobe.com ✓	73434

1

*

```
mysql> select * from products;
+----+----------------+--------+-----+
| id | name           | price  | qty |
+----+----------------+--------+-----+
|  1 | iPhone 16      |  89000 | 100 |
|  2 | Sony Bravia    | 297000 | 100 |
|  3 | Wacom          |   5350 | 100 |
|  4 | Logitech Mouse |    900 | 100 |
|  5 | LG AC          |  45000 | 100 |
+----+----------------+--------+-----+
```

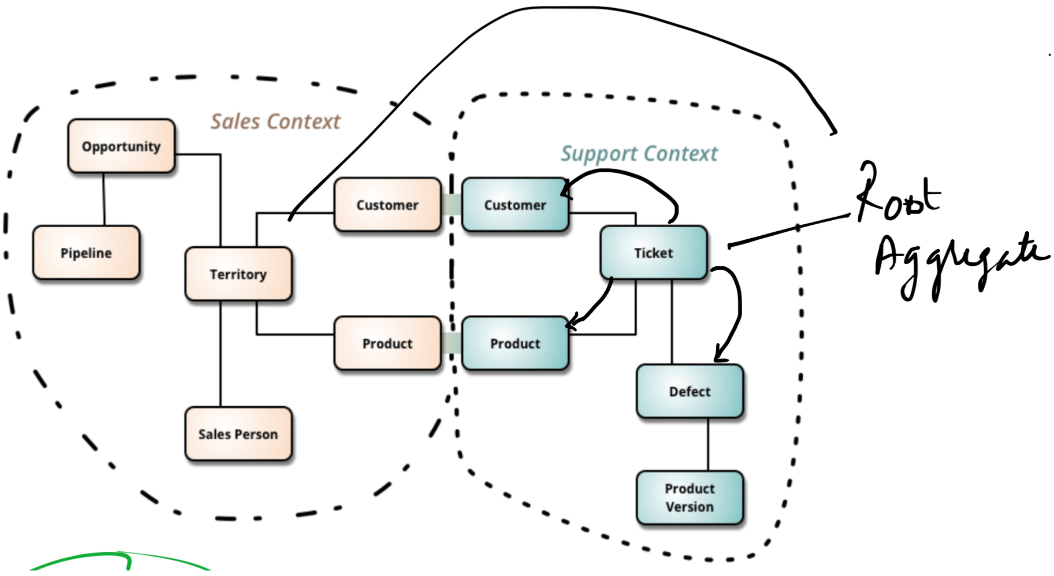
1

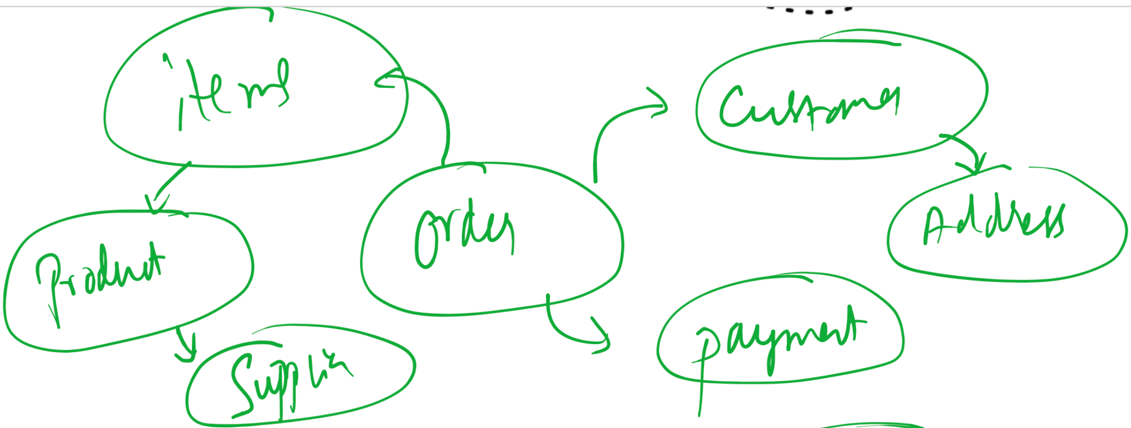
*

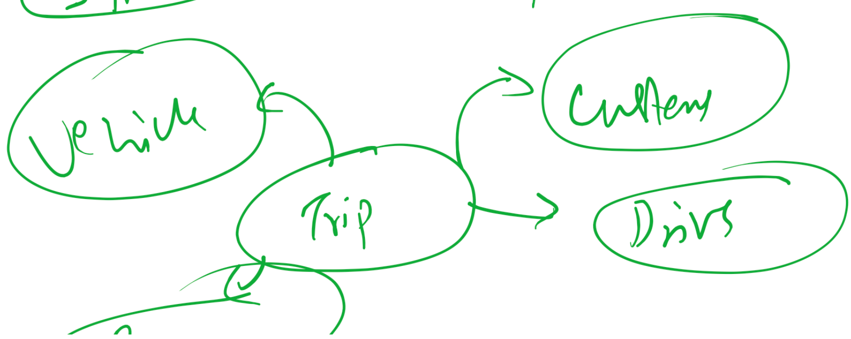
Line_items

Item_id	order_fk	Product_fk	qty	amount
612	343 ✓	2	1	<<>>
613	343 ✓	1	2	<<>>
614	344	1	1	<<>>
615	344	4	3	<<>>

Cardinality from Java's perspective


Sales Context
Opportunity
Pipeline
Territory
Customer
Product
Sales Person
Support Context
Customer
Ticket
Product
Defect
Product Version
Root Aggregate


items
Product
Supplier
Order
Customer
Address
payment


Vehicle
Trip
Customer
Drivers